

A sum-of-squares proof of Ulas's real-zero conjecture for Stern polynomials

Root localization, congruences, and an irreducible subfamily

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Abstract

For the Stern polynomials defined by $B_0 = 0$, $B_1 = 1$, $B_{2m} = tB_m$ and $B_{2m+1} = B_m + B_{m+1}$, Ulas formulated a conjecture that B_{h_n} has no real zeros, where $h_n = \frac{2}{3}(4^n - 1)(2 \cdot 4^n + 1) + 1$. We prove this statement by exhibiting an exact positive-definite quadratic form in two consecutive auxiliary polynomials. The identity is obtained independently from the defining Stern recurrence, and the argument includes the exceptional parameter $t = 0$. A tridiagonal-matrix representation then places every zero strictly to the left of $\operatorname{Re} t = -1/4$. Further consequences are a rational generating function, an exact formula for the degree after reduction modulo three, and irreducibility over $\mathbb{Q}$ whenever $2n + 1$ is a power of three. For each fixed positive integer j , we also construct a conjugate pair of zeros with real part of order $-n^2/(j^2\pi^2)$ and imaginary part of order $\pm\sqrt{3}n/(j^2\pi^2)$. All universal claims are proved analytically. A separate, dependency-free program supplies 6,722 exact checks; numerical illustrations are explicitly non-certifying.

Scope and priority. The target is Conjecture 4.4 in the accessible version arXiv:1909.10844v1 of Ulas's paper [2]. Targeted literature searches did not identify a subsequent resolution. The journal PDF was not accessible through the available subscription route, so the conjecture's status in the final published text was not certified. This note proves the precisely stated mathematical assertion; it does *not* claim independently verified publication priority or a complete bibliographic novelty check. It has not been externally peer-reviewed or formalized in a proof assistant.

1 The target and the outcome

Stern polynomials are a natural combinatorial family: the coefficient of t^k in $B_m(t)$ counts hyperbinary representations of $m - 1$ containing exactly k digits equal to one [1]. Equivalently, their generating function, interpreted as a formal power series in x , is

$$\sum_{m \geq 0} B_m(t) x^m = x \prod_{r \geq 0} (1 + tx^{2^r} + x^{2^{r+1}}). \quad (1.1)$$

Thus a zero-location question for B_m is a question about a counting polynomial, rather than an arbitrarily engineered polynomial family. The product also follows directly by independently choosing the digit 0, 1, or 2 at each binary position.

Throughout the note, the defining recurrence is

$$B_0 = 0, \quad B_1 = 1, \quad B_{2m} = tB_m, \quad B_{2m+1} = B_m + B_{m+1}. \quad (1.2)$$

For the last identity, $m = 0$ is interpreted as the valid identity $B_1 = B_0 + B_1$. Set

$$a_n = \frac{4^n - 1}{3}, \quad h_n = \frac{2}{3}(4^n - 1)(2 \cdot 4^n + 1) + 1, \quad A_n = B_{a_n}, \quad W_n = B_{h_n} \quad (n \geq 0). \quad (1.3)$$

The sequence (h_n) begins

$$1, 19, 331, 5419, 87211, 1397419, \dots$$

The conjectural assertion selected for attack was that W_n has no real zeros for every $n \geq 0$ [2, Conjecture 4.4]. The outcome is a proof, not a counterexample. Its entire mechanism is the identity

$$W_n(t) = \left(A_{n+1}(t) - \frac{t}{2} A_n(t) \right)^2 + \frac{3}{4} t^2 A_n(t)^2. \quad (1.4)$$

The only remaining issue is to exclude simultaneous vanishing of the two squares. A second-order recurrence does that immediately.

The auxiliary quadratic expression in adjacent Stern polynomials used below is already present in Ulas's proof of Theorem 3.1 [2]. The contribution here is the positivity argument and its explicitly proved consequences, not the discovery of that adjacent-polynomial expression. A few inconsistent formulas in the arXiv version make an independent derivation worthwhile; [section A](#) records the corrections relevant to this note.

2 A self-contained derivation of the certificate

2.1 Binary concatenation

We first prove the standard Stern concatenation identity; it is also recorded in [2, Lemma 2.1].

Lemma 2.1. *For integers $k, m \geq 0$ and $0 \leq r \leq 2^k$,*

$$B_{2^k m + r} = B_{2^{k-r}} B_m + B_r B_{m+1}. \quad (2.1)$$

Proof. When $k = 0$, the two possibilities $r = 0, 1$ give identities. Suppose the formula holds for $2^k = q$ and consider $2q$. If $r = 2s$, multiply the identity for $qm + s$ by t :

$$B_{2qm+2s} = t B_{qm+s} = B_{2q-2s} B_m + B_{2s} B_{m+1}.$$

If $r = 2s + 1$, where $0 \leq s < q$, add the identities for $qm + s$ and $qm + s + 1$. The coefficients become

$$B_{q-s} + B_{q-s-1} = B_{2q-2s-1}, \quad B_s + B_{s+1} = B_{2s+1}.$$

This is the required formula. The endpoint $r = 2q$ was included in the even case, completing the induction. $\square$

2.2 The two-dimensional recurrence

For clarity, distinguish the adjacent polynomial

$$D_n = B_{a_n+1}$$

from $A_{n+1} = B_{a_{n+1}}$. Since $a_{n+1} = 4a_n + 1$, the defining recurrence gives

$$\begin{pmatrix} A_{n+1} \\ D_{n+1} \end{pmatrix} = \begin{pmatrix} t+1 & 1 \\ t & t \end{pmatrix} \begin{pmatrix} A_n \\ D_n \end{pmatrix}, \quad (A_0, D_0) = (0, 1). \quad (2.2)$$

In particular,

$$A_0 = 0, \quad A_1 = 1, \quad A_{n+2} = (1+2t)A_{n+1} - t^2A_n, \quad D_n = A_{n+1} - (t+1)A_n. \quad (2.3)$$

For example, the scalar recurrence follows by substituting $D_{n+1} = t(A_n + D_n)$ into $A_{n+2} = (t+1)A_{n+1} + D_{n+1}$.

Lemma 2.2. *If $t \neq 0$, then $A_n(t)$ and $A_{n+1}(t)$ cannot both be zero. Moreover, $A_n(0) = 1$ for $n \geq 1$.*

Proof. A simultaneous zero at $t \neq 0$ propagates backward through (2.3): $A_{n-1}(t) = 0$, then $A_{n-2}(t) = 0$, and eventually $A_1(t) = 0$, contradicting $A_1 = 1$. For $n = 0$ the contradiction is immediate. Setting $t = 0$ in the recurrence proves the second assertion from $A_1(0) = 1$. $\square$

2.3 The exact quadratic identity

Proposition 2.3. *For every $n \geq 0$, the following identities hold in $\mathbb{Z}[t]$:*

$$W_n = (t^2 + t + 1)A_n^2 + (t + 2)A_nD_n + D_n^2, \quad (2.4)$$

$$W_n = A_{n+1}^2 - tA_nA_{n+1} + t^2A_n^2. \quad (2.5)$$

Proof. Put $m = a_n$ and $q = 4^n = 3m + 1$. A direct calculation gives

$$h_n = 4qm + 2m + 1, \quad 4q - 2m - 1 = 10m + 3.$$

Use [theorem 2.1](#), with the power of two $4q$, to obtain

$$W_n = B_{10m+3}B_m + B_{2m+1}B_{m+1}. \quad (2.6)$$

Two more applications, now with the power of two q , give

$$\begin{aligned} B_{5m+1} &= B_{q+2m} = B_{m+1} + tB_{2m} = D_n + t^2A_n, \\ B_{5m+2} &= B_{q+2m+1} = B_m + tB_{2m+1} = (1+t)A_n + tD_n. \end{aligned}$$

Here $0 \leq 2m \leq 2m+1 \leq q$, so the concatenation hypotheses include every case, even $n = 0$. Adding these identities yields

$$B_{10m+3} = B_{5m+1} + B_{5m+2} = (t^2 + t + 1)A_n + (t + 1)D_n.$$

Since $B_{2m+1} = A_n + D_n$, substitution into (2.6) proves (2.4). Finally replace D_n by $A_{n+1} - (t+1)A_n$. The coefficients of A_n^2 and A_nA_{n+1} simplify to t^2 and $-t$, respectively, which proves (2.5). $\square$

3 Proof of the real-zero conjecture

Theorem 3.1. *For every integer $n \geq 0$ and every real number t ,*

$$W_n(t) > 0.$$

Consequently, Conjecture 4.4 in [arXiv:1909.10844v1](#) holds as stated.

Proof. Completing the square in (2.5) gives

$$W_n(t) = \left(A_{n+1}(t) - \frac{t}{2} A_n(t) \right)^2 + \frac{3}{4} t^2 A_n(t)^2.$$

For real t both terms are nonnegative. If $t \neq 0$ and their sum were zero, the second term would force $A_n(t) = 0$ and the first would then force $A_{n+1}(t) = 0$, contradicting [theorem 2.2](#). At $t = 0$, one has $W_n(0) = A_{n+1}(0)^2 = 1$. This covers $n = 0$ as well, where $W_0 = 1$. $\square$

The proof is independent of numerical root finding, coefficient truncation, asymptotic estimates, or the sign of t . In particular, it does not infer strict positivity merely from a nonnegative expression: the common-zero obstruction and $t = 0$ have both been treated.

It is useful to retain an integer-coefficient version of the certificate:

$$4W_n = (2A_{n+1} - tA_n)^2 + 3(tA_n)^2. \quad (3.1)$$

Over $\mathbb{Q}(\sqrt{-3})$ the same identity becomes a norm factorization. Writing

$$\zeta = \frac{1 + i\sqrt{3}}{2}, \quad \zeta + \bar{\zeta} = 1, \quad \zeta \bar{\zeta} = 1, \quad (3.2)$$

we have

$$W_n = (A_{n+1} - \zeta t A_n)(A_{n+1} - \bar{\zeta} t A_n). \quad (3.3)$$

This factorization is the entry point for the stronger zero-location theorem.

4 An explicit enclosure for every complex zero

Define monic polynomials $C_j(s)$ by

$$C_{-1} = 0, \quad C_0 = 1, \quad C_{j+1} = sC_j - C_{j-1} \quad (j \geq 0). \quad (4.1)$$

These are rescaled Chebyshev polynomials of the second kind; the recurrence is all that is needed here. For $t \neq 0$, let $s = 2 + 1/t$. Comparing (2.3) and (4.1) gives

$$A_n(t) = t^{n-1} C_{n-1}(s) \quad (n \geq 1), \quad A_{n+1}(t) = t^n C_n(s). \quad (4.2)$$

Thus the two factors of W_n correspond to

$$C_n(s) - \zeta C_{n-1}(s), \quad C_n(s) - \bar{\zeta} C_{n-1}(s).$$

Theorem 4.1. *Let $n \geq 1$ and put*

$$\rho_n = 2 \cos \frac{\pi}{2n+1} < 2.$$

Every zero t of W_n satisfies

$$\left| 2 + \frac{1}{t} \right| \leq \rho_n. \quad (4.3)$$

Equivalently, all zeros belong to the closed disk

$$\left| t + \frac{2}{4 - \rho_n^2} \right| \leq \frac{\rho_n}{4 - \rho_n^2}. \quad (4.4)$$

In particular,

$$\boxed{\operatorname{Re} t \leq -\frac{1}{2 + \rho_n} < -\frac{1}{4}, \quad |t| \leq \frac{1}{2 - \rho_n}.} \quad (4.5)$$

There are exactly n zeros in each open half-plane $\operatorname{Im} t > 0$ and $\operatorname{Im} t < 0$, counted with algebraic multiplicity.

Proof. Let $J_n(\zeta)$ be the $n \times n$ tridiagonal matrix whose upper and lower off-diagonal entries are one, with diagonal $(0, \dots, 0, \zeta)$. For $n = 1$ this means $J_1(\zeta) = [\zeta]$. Expansion of the determinant along the last row, together with (4.1), proves

$$\det(sI - J_n(\zeta)) = C_n(s) - \zeta C_{n-1}(s). \quad (4.6)$$

For example, the last step of the determinant recurrence is $(s - \zeta)C_{n-1} - C_{n-2} = C_n - \zeta C_{n-1}$.

Suppose $J_n(\zeta)v = sv$ with $v \neq 0$. The last component v_n is nonzero: otherwise the last row forces $v_{n-1} = 0$, and backward substitution forces every component to vanish. Taking the imaginary part of $v^* J_n(\zeta)v = s v^* v$ gives

$$\operatorname{Im} s = \frac{\sqrt{3} |v_n|^2}{2 \|v\|^2} > 0, \quad (4.7)$$

because the off-diagonal part is real symmetric. Since $t = 1/(s - 2)$, the corresponding t lies in the lower half-plane. Conjugating the argument gives the upper half-plane for the other factor.

For the modulus estimate, the entrywise absolute-value matrix is $|J_n(\zeta)| = H_n := J_n(1)$. Put $\theta = \pi/(2n+1)$ and $w_j = \sin(j\theta)$ for $1 \leq j \leq n$. All w_j are positive, and

$$H_n w = 2 \cos \theta w = \rho_n w.$$

The interior rows use the sine addition formula; the last row uses $\sin((n+1)\theta) = \sin(n\theta)$, and the first row uses $\sin 0 = 0$. The same verification covers $n = 1$. Choose an index i maximizing $|v_i|/w_i$, and denote that positive maximum by M . The triangle inequality now gives

$$|s| |v_i| \leq \sum_j (H_n)_{ij} |v_j| \leq M \sum_j (H_n)_{ij} w_j = M \rho_n w_i = \rho_n |v_i|.$$

Hence $|s| \leq \rho_n$. The proof applies without change to $J_n(\bar{\zeta})$, establishing (4.3) for every zero.

Writing $t = x + iy$, inequality (4.3) is

$$(4 - \rho_n^2)(x^2 + y^2) + 4x + 1 \leq 0.$$

Completing this square proves (4.4), whose rightmost real coordinate is $-1/(2 + \rho_n)$. The modulus estimate follows from $|s - 2| \geq 2 - |s| \geq 2 - \rho_n$.

Finally, theorem 5.1 below proves that A_n has leading coefficient n and degree $n - 1$. Each factor in (3.3) therefore has degree n , with nonzero leading coefficient $(n + 1) - \zeta n$ or its conjugate. The sign computation (4.7) assigns all n zeros of each factor, with multiplicity, to the corresponding half-plane. $\square$

No simplicity assertion is needed in this argument. The enclosure concerns the special family W_n only, not all Stern polynomials. In particular, it does not settle the general half-plane conjecture discussed in [3].

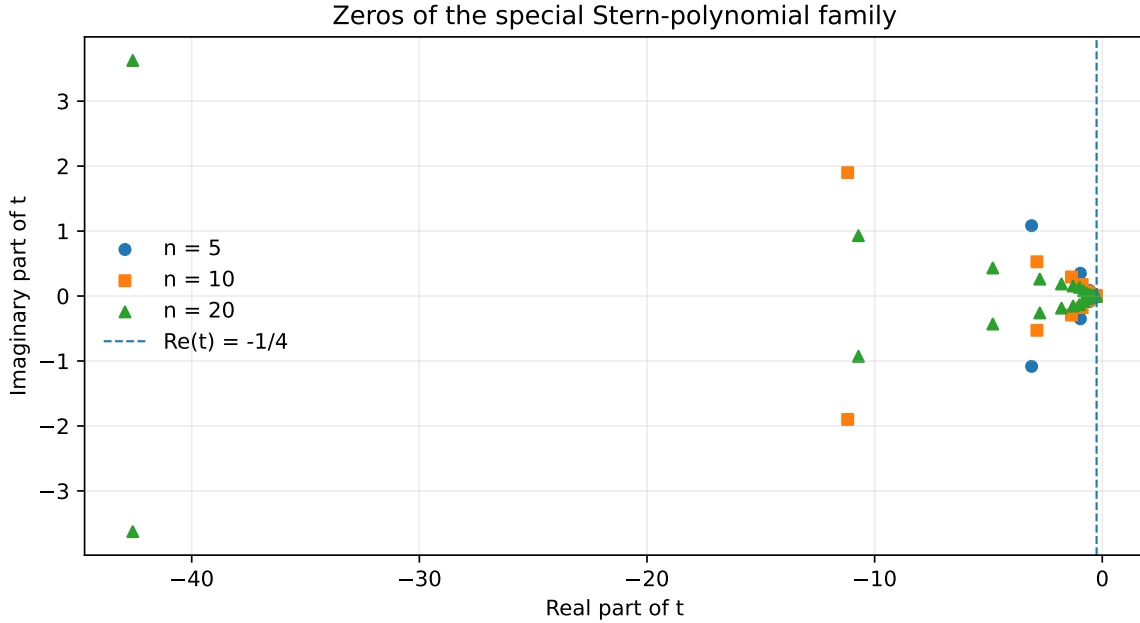


Figure 1: Numerical illustrations for $n = 5, 10, 20$, obtained from the $n \times n$ tridiagonal matrices and their conjugates. Points close to the real axis may appear to lie on it at this scale; theorem 3.1 excludes that possibility. The dashed line is $\text{Re } t = -1/4$. Neither the plot nor floating-point eigenvalues are used in the proofs.

5 Generating functions and arithmetic consequences

5.1 Coefficient formulas and a rational generating function

Proposition 5.1. *For $n \geq 1$,*

$$A_n(t) = \sum_{j=0}^{n-1} \binom{2n-1-j}{j} t^j. \quad (5.1)$$

Consequently,

$$\deg W_n = 2n, \quad \text{lc}(W_n) = n^2 + n + 1, \quad W_n(0) = 1. \quad (5.2)$$

The degree and leading-coefficient formulas also hold at $n = 0$.

Proof. Introduce Fibonacci polynomials $F_0 = 0$, $F_1 = 1$ and $F_{m+2} = F_{m+1} + tF_m$. Pascal's identity proves by induction that

$$F_m(t) = \sum_{j=0}^{\lfloor (m-1)/2 \rfloor} \binom{m-1-j}{j} t^j \quad (m \geq 1). \quad (5.3)$$

Moreover,

$$F_{m+4} = (1+t)F_{m+2} + tF_{m+1} = (1+2t)F_{m+2} - t^2F_m.$$

The subsequence F_{2n} therefore satisfies the same recurrence and initial values as A_n , so $A_n = F_{2n}$ and (5.1) follows. Its leading coefficient is $\binom{n}{n-1} = n$. The degree- $2n$ coefficient of (2.5) is

$$(n+1)^2 - n(n+1) + n^2 = n^2 + n + 1 > 0.$$

The constant coefficient and the case $n = 0$ are immediate. □

There is also a useful nonrecursive square-minus-monomial formula.

Proposition 5.2. *For every $n \geq 0$,*

$$W_n(t) = (1+t)F_{2n+1}(t)^2 - t^{2n+1}. \quad (5.4)$$

In particular, $W_n(-1) = 1$.

Proof. Cassini's identity for these Fibonacci polynomials is

$$F_{m+1}F_{m-1} - F_m^2 = (-1)^m t^{m-1} \quad (m \geq 1).$$

For completeness, its left side at $m = 1$ is -1 , and replacing m by $m+1$ multiplies it by $-t$, as follows by substituting $F_{m+2} = F_{m+1} + tF_m$ and $F_{m+1} - F_m = tF_{m-1}$. Take $m = 2n+1$ and set $X = F_{m+1} = A_{n+1}$, $Y = tF_{m-1} = tA_n$. Then $X - Y = F_m$ and $XY = tF_m^2 - t^m$ by Cassini. Equation (2.5) becomes

$$W_n = X^2 - XY + Y^2 = (X - Y)^2 + XY = (1+t)F_m^2 - t^m.$$

Substituting $t = -1$ finishes the proof. □

Let u be a second formal variable. The generating function across the selected Stern indices is particularly small:

Theorem 5.3. *In $\mathbb{Z}[t][[u]]$,*

$$\sum_{n \geq 0} W_n(t) u^n = \frac{1 - tu + t^4 u^2}{1 - (3t^2 + 4t + 1)u + t^2(3t^2 + 4t + 1)u^2 - t^6 u^3}. \quad (5.5)$$

The denominator factors as

$$(1 - t^2 u)(1 - (2t^2 + 4t + 1)u + t^4 u^2).$$

Proof. Put $a = 1 + 2t$ and consider

$$V_n = \begin{pmatrix} A_{n+1}^2 \\ A_{n+1}A_n \\ A_n^2 \end{pmatrix}, \quad V_{n+1} = SV_n, \quad S = \begin{pmatrix} a^2 & -2at^2 & t^4 \\ a & -t^2 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

The characteristic polynomial of S is

$$(\lambda - t^2)(\lambda^2 - (2t^2 + 4t + 1)\lambda + t^4).$$

Cayley–Hamilton, followed by the linear functional $(1, -t, t^2)$, therefore gives, with $p = 3t^2 + 4t + 1$,

$$W_n = pW_{n-1} - t^2pW_{n-2} + t^6W_{n-3} \quad (n \geq 3). \quad (5.6)$$

This recurrence also appears in [2]; the argument here checks it without using the initial polynomials printed there. Direct computation from (2.5) gives

$$\begin{aligned} W_0 &= 1, \\ W_1 &= 1 + 3t + 3t^2, \\ W_2 &= 1 + 7t + 17t^2 + 17t^3 + 7t^4. \end{aligned}$$

Multiplication of the formal series by the denominator in (5.5) leaves the coefficients 1, $W_1 - pW_0 = -t$, and $W_2 - pW_1 + t^2pW_0 = t^4$ in degrees zero, one, and two; all higher coefficients vanish by (5.6). This proves the formula as a formal identity, including all exceptional values of t . $\square$

5.2 Exact reduction modulo three

Define Lucas polynomials by

$$L_0 = 2, \quad L_1 = 1, \quad L_{m+2} = L_{m+1} + tL_m.$$

For $m \geq 1$, the recurrence and initial values show that

$$L_m = F_{m+1} + tF_{m-1}. \quad (5.7)$$

For odd $m = 2k + 1$, this polynomial has degree k and leading coefficient $2k + 1 = m$: the two top coefficients on the right side are $k + 1$ and k . This includes $m = 1$.

Write $\bar{P}$ for the reduction of $P \in \mathbb{Z}[t]$ in $\mathbb{F}_3[t]$.

Theorem 5.4. *Let $n \geq 0$, and write $2n + 1 = 3^r v$ with $3 \nmid v$. Then*

$$\overline{W_n} = \overline{L_v}^{2 \cdot 3^r}, \quad \deg \overline{W_n} = 2n + 1 - 3^r. \quad (5.8)$$

The reduced polynomial is monic. In particular,

$$\boxed{W_n \equiv 1 \pmod{3} \iff 2n + 1 \text{ is a power of } 3.} \quad (5.9)$$

Here $3^0 = 1$ is allowed.

Proof. In characteristic three, $-1 = 2$. Thus (2.5) and (5.7) give

$$\overline{W_n} = (\overline{A_{n+1}} + t\overline{A_n})^2 = \overline{L_{2n+1}}^2. \quad (5.10)$$

The polynomial identity

$$L_{3m} = L_m^3 - 3(-t)^m L_m \quad (5.11)$$

follows by writing $L_m = \alpha^m + \beta^m$ in the splitting field of $X^2 - X - t$, where $\alpha\beta = -t$, and expanding the cube. Both sides are integer polynomials, so it is an identity in $\mathbb{Z}[t]$. Modulo three it gives $\overline{L_{3m}} = \overline{L_m}^3$. Iteration and (5.10) prove the first formula in (5.8).

Since v is odd and not divisible by three, the top coefficient v of L_v survives reduction. Therefore

$$\deg \overline{W_n} = 2 \cdot 3^r \frac{v-1}{2} = 3^r(v-1) = 2n+1-3^r.$$

Its leading coefficient is $v^{2 \cdot 3^r} = 1$ in $\mathbb{F}_3$. The reduced polynomial is constant exactly when $v = 1$, and in that case $L_v = L_1 = 1$. This proves (5.9). $\square$

The forward construction $n = (3^r - 1)/2 \Rightarrow W_n \equiv 1 \pmod{3}$ is a result of Ulas [2, Theorem 3.1]. The argument above independently proves it and also supplies the converse and the exact degree drop within this family.

5.3 An infinite irreducible subsequence

Corollary 5.5. *For every $r \geq 1$, with $n = (3^r - 1)/2$, the polynomial W_n is irreducible in $\mathbb{Q}[t]$ and has degree $3^r - 1$.*

Proof. By theorem 5.4, every nonconstant coefficient of W_n is divisible by three, whereas its constant coefficient is one. The leading coefficient from theorem 5.1 is

$$n^2 + n + 1 = \frac{3^{2r} + 3}{4} = 3 \frac{3^{2r-1} + 1}{4}.$$

It is divisible by three but not nine, since $3^{2r-1} + 1 \equiv 1 \pmod{3}$ and 4 is a unit modulo three. The reciprocal polynomial

$$R_n(t) = t^{2n} W_n(1/t)$$

is therefore monic, all its nonleading coefficients are divisible by three, and its constant coefficient is not divisible by nine. Eisenstein's criterion at three makes R_n irreducible over $\mathbb{Q}$. Because $W_n(0) = 1$, taking reciprocals carries any nontrivial factorization of W_n to a nontrivial factorization of R_n ; hence W_n is irreducible.

For completeness, the relevant Eisenstein argument is elementary. By Gauss's lemma, a rational factorization of a monic integer polynomial can be taken monic with integer coefficients. Reduction of R_n modulo three is t^{2n} , so the reductions of two positive-degree monic factors would both be powers of t . Both constant coefficients would be divisible by three, forcing the constant coefficient of R_n to be divisible by nine, a contradiction. $\square$

This does not assert irreducibility for every n . The indices $n = 1, 4, 13, 40, 121, 364, \dots$ form the proven subsequence.

6 Asymptotic nonreal zeros

The matrix enclosure permits $|t|$ of order n^2 . We next show that this scale actually occurs, with imaginary parts of order n . The unboundedness of imaginary parts across all Stern polynomials is already known [3, Theorem 10]; the result here is an explicit asymptotic for the particular family selected in section 1.

Theorem 6.1. *Fix an integer $j \geq 1$. For all sufficiently large n , there is a zero $t_{n,j}$ of W_n in the lower half-plane such that, with $\zeta = (1 + i\sqrt{3})/2$,*

$$t_{n,j} = -\frac{(n + \zeta)^2}{j^2\pi^2} - \frac{1}{12} - \frac{i\sqrt{3}}{3(n + \zeta)} + O_j(n^{-2}). \quad (6.1)$$

Its conjugate is also a zero. In particular,

$$\operatorname{Re} t_{n,j} = -\frac{n^2 + n - \frac{1}{2}}{j^2\pi^2} - \frac{1}{12} + O_j(n^{-2}), \quad (6.2)$$

$$\operatorname{Im} t_{n,j} = -\frac{\sqrt{3}(n + \frac{1}{2})}{j^2\pi^2} + O_j(n^{-1}), \quad (6.3)$$

$$\frac{\operatorname{Re} t_{n,j}}{(\operatorname{Im} t_{n,j})^2} \rightarrow -\frac{j^2\pi^2}{3}. \quad (6.4)$$

The notation O_j allows constants depending on the fixed integer j ; no estimate uniform in $1 \leq j \leq n$ is asserted.

Proof. From (4.1), or directly from the sine addition formula,

$$C_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}$$

whenever $\sin \theta \neq 0$. The factor $C_n - \zeta C_{n-1}$ therefore vanishes if

$$\sin((n+1)\theta) - \zeta \sin(n\theta) = 0. \quad (6.5)$$

Near zero, define the analytic function

$$\delta(\theta) = \arctan\left(\frac{\sin \theta}{\cos \theta - \zeta}\right), \quad \delta(0) = 0, \quad (6.6)$$

using the local branch of $\arctan$ at zero. The denominator is nonzero there because $1 - \zeta \neq 0$. By the sine addition formula, (6.5) is equivalent, in a sufficiently small neighborhood, to

$$\sin(n\theta + \delta(\theta)) = 0.$$

Indeed, multiplying its left side by $(\cos \theta - \zeta)/\cos \delta(\theta)$ gives exactly the left side of (6.5); the multiplier is nonzero near zero.

To construct the required root rigorously, put $\varepsilon = 1/n$ and $\theta = \varepsilon u$. The equation

$$u + \delta(\varepsilon u) = j\pi$$

has a unique analytic solution $u = u(\varepsilon)$ near $(u, \varepsilon) = (j\pi, 0)$, by the analytic implicit function theorem: the derivative with respect to u there is one. For positive integers n sufficiently large, $\theta_n = u(1/n)/n$ is small and nonzero, so $\sin \theta_n \neq 0$. Consequently (6.5) gives an actual zero of the polynomial factor, not a spurious zero caused by the sine denominator.

We now compute the expansion. The identities $(1 - \zeta)^{-1} = \zeta$, $\zeta^2 = \zeta - 1$, and $\zeta^3 = -1$ give

$$\begin{aligned}\delta(\theta) &= \zeta\theta + \left(\frac{\zeta^2}{2} - \frac{\zeta}{6} - \frac{\zeta^3}{3}\right)\theta^3 + O(\theta^5) \\ &= \zeta\theta + \frac{i\sqrt{3}}{6}\theta^3 + O(\theta^5).\end{aligned}$$

Write $N = n + \zeta$, $q = j\pi$, and $c = i\sqrt{3}/6$. The equation $n\theta_n + \delta(\theta_n) = q$, together with $\theta_n = O_j(n^{-1})$, now yields

$$\theta_n = \frac{q}{N} - \frac{cq^3}{N^4} + O_j(n^{-6}), \quad \frac{1}{\theta_n^2} = \frac{N^2}{q^2} + \frac{2c}{N} + O_j(n^{-3}).$$

For example, the first estimate follows by first obtaining $\theta_n = q/N + O_j(n^{-4})$ and substituting this back into the cubic term; the remainder $O(\theta_n^5)$ contributes $O_j(n^{-6})$ after division by N .

The corresponding zero in the original variable is

$$t_{n,j} = \frac{1}{2\cos \theta_n - 2} = -\frac{1}{4\sin^2(\theta_n/2)} = -\frac{1}{\theta_n^2} - \frac{1}{12} + O(\theta_n^2).$$

Substitution proves (6.1). Its imaginary part is negative for sufficiently large n , consistently with theorem 4.1. Since W_n has real coefficients, the conjugate is a zero as well. Taking real and imaginary parts gives (6.2) and (6.3); the real part of $-i\sqrt{3}/(3N)$ is $O(n^{-2})$. Finally, division of the leading terms gives (6.4). $\square$

The asymptotic is more informative than the mere existence of nonreal zeros. Along these branches the distance from the real axis grows unboundedly, even though the angle to the negative real axis tends to zero. It also shows why a search restricted to a fixed rectangular window in the complex plane would miss some of the most distinctive roots.

7 Computational audit and reproducibility

The universal results in this note rest on the proofs, not on a finite search. The programs are included to make indexing mistakes, coefficient errors, and numerical overinterpretation easier to detect. The primary exact evaluator constructs (B_m, B_{m+1}) directly from the binary digits of m :

$$(P, Q) \xrightarrow{0} (tP, P + Q), \quad (P, Q) \xrightarrow{1} (P + Q, tQ), \quad (P, Q) = (0, 1) \text{ initially.}$$

This route is independent of the second-order recurrence for A_n and of the quadratic formula for W_n . The latter formulas are tested against it, not used to define the purported ground truth.

7.1 Exact checks actually executed

The delivered code/verify_exact.py uses only the Python standard library, integer coefficient arrays, and rational Sturm remainders. The recorded run used Python 3.13.5 and passed 6,722

checks. The scope is summarized in [table 1](#). Counts refer to individual assertions, not to distinct conjectures or independently formalized proofs.

Check	Executed range or extent
Binary evaluator against the direct Stern recurrence	Indices $0 \leq m \leq 1024$
Hyperbinary generating product against B_{m+1}	Coefficients $0 \leq m \leq 64$ in x
Concatenation, including both endpoints	$0 \leq k \leq 6, 0 \leq m \leq 24, 0 \leq r \leq 2^k$
Three constructions of A_n ; adjacent conversion	$0 \leq n \leq 160$
Quadratic, sum-of-squares, and square-minus-monomial identities	$0 \leq n \leq 160$
Degree, leading coefficient, reduction modulo three	$0 \leq n \leq 160$
Third-order recurrence and generating-function numerator	$3 \leq n \leq 160$; initial two coefficients
Reciprocal Eisenstein conditions	$r = 1, \dots, 6$; maximum $n = 364$, degree 728
Exact rational Sturm counts, with control examples	$n = 1, \dots, 12$; zero real roots in every case

Table 1: The exact finite audit. Full assertion counts and sample coefficients are stored in `results/exact_verification.json`.

The largest Eisenstein case was also checked against a direct Stern-index evaluation, rather than only against the abstract congruence criterion. Sturm counts are exact over $\mathbb{Q}$; they do not rely on small imaginary parts returned by a floating-point root finder.

7.2 Numerical illustrations

The optional script code `numerical_checks.py` computes matrix eigenvalues for seven values of n up to 100, and tests the displayed spectral inequalities numerically. It also solves the sine equation at 80 decimal digits for $j = 1, 2$ and $n = 10, 20, 50, 100, 200, 500$. Each result is cross-checked by evaluating the polynomial recurrence $C_n - \zeta C_{n-1}$, with the normalized residual required to be less than 10^{-65} . This is a numerical diagnostic, not interval-certified root isolation.

Let

$$T_{n,1}^{(0)} = -\frac{(n + \zeta)^2}{\pi^2} - \frac{1}{12}.$$

For the lower root with $j = 1$, the observed values in [table 2](#) illustrate the proven first correction: $n|t_{n,1} - T_{n,1}^{(0)}| \rightarrow \sqrt{3}/3$. This limit itself follows from (6.1); it is not inferred from the table.

n	$\operatorname{Re} t_{n,1}$	$\operatorname{Im} t_{n,1}$	$n t_{n,1} - T_{n,1}^{(0)} $
10	-11.18381641	-1.89989755	0.57511034
20	-42.58891626	-3.62608332	0.56999317
50	-258.40190478	-8.87387171	0.57275425
100	-1023.37668120	-17.64283788	0.57476199
200	-4073.14426863	-35.18931419	0.57598204
500	-25380.98917731	-87.83561926	0.57678499

Table 2: Rounded numerical data. The limiting value in the last column is $\sqrt{3}/3 \approx 0.57735027$. Higher precision and the refined-error columns are in `results/root_asymptotics.csv`.

7.3 How to reproduce the archive

From the extracted archive directory, run

```
python3 code/verify_exact.py
pdflatex -interaction=nonstopmode -halt-on-error stern_conjecture_note.tex
pdflatex -interaction=nonstopmode -halt-on-error stern_conjecture_note.tex
```

The optional numerical script requires `mpmath`, `numpy`, and `matplotlib`. Installation details are in `requirements-optional.txt`. The figure PDF is included, so rebuilding the note does not require these optional packages. No program needs a network connection.

8 What has and has not been established

The selected assertion is proved for every allowed index and every real argument, with no numerical hypothesis. The stronger complex-zero localization, the generating function, the exact reduction-modulo-three criterion, the infinite irreducible subsequence, and the fixed- j root asymptotic likewise have unconditional proofs within this note.

There are separate limits to the claim. The targeted literature search is not a proof of priority, and it did not verify the final journal version of the 2019 conjecture. The current write-up has not received external mathematical review. The code provides a finite audit, not a formal proof of the universal statements. Finally, none of the arguments here resolves the zero geometry or irreducibility of *all* Stern polynomials.

The central mathematical lesson is a specific structural one. The sparse binary indices h_n lead to a quadratic form in two nearby auxiliary polynomials. Once that form is written correctly, its discriminant is $-3t^2$, and the real-zero question reduces to a one-line backward-recursion argument. The same form then exposes both an Eisenstein norm and a tridiagonal boundary perturbation, explaining why arithmetic and spectral consequences emerge from the same certificate.

A Source audit for the arXiv version

The following corrections concern only arXiv:1909.10844v1, whose PDF pages 13, 14, and 16 were visually checked. They are not assertions about the final journal typesetting.

On page 13, the adjacent-polynomial identity is consistent with (2.4). After the displayed substitution for B_{a_n+1} , however, the quadratic in $(X, Y) = (A_n, A_{n+1})$ must be

$$F(X, Y) = t^2 X^2 - tXY + Y^2,$$

not the expression printed there. On page 14, the initial values compatible with the Stern definition are

$$W_1 = 3t^2 + 3t + 1, \quad W_2 = 7t^4 + 17t^3 + 17t^2 + 7t + 1.$$

The coefficients of t in W_1 and t^2 in W_2 differ from those printed in the arXiv recurrence display. The third-order recurrence itself is consistent with the independent derivation in [theorem 5.3](#).

These observations are checks on the formulas actually used, not an attempt to audit every statement of the source. The proof in this note starts from (1.2) and does not depend on adopting any corrected formula on authority.

B Small exact examples

The first auxiliary polynomials are

$$\begin{aligned} A_0 &= 0, & A_1 &= 1, & A_2 &= 1 + 2t, \\ A_3 &= 1 + 4t + 3t^2, & A_4 &= 1 + 6t + 10t^2 + 4t^3. \end{aligned}$$

The first nonconstant members of the target family are

$$\begin{aligned} W_1 &= 1 + 3t + 3t^2, \\ W_2 &= 1 + 7t + 17t^2 + 17t^3 + 7t^4, \\ W_3 &= 1 + 11t + 47t^2 + 99t^3 + 108t^4 + 58t^5 + 13t^6, \\ W_4 &= 1 + 15t + 93t^2 + 309t^3 + 597t^4 \\ &\quad + 681t^5 + 444t^6 + 150t^7 + 21t^8. \end{aligned}$$

For example,

$$W_1 = (1 + 2t)^2 - t(1 + 2t) + t^2 = 1 + 3t + 3t^2,$$

and its zeros are $-1/2 \pm i\sqrt{3}/6$. The cases $n = 1$ and $n = 4$ have $2n + 1 = 3, 9$; every nonconstant coefficient is divisible by three, and their leading coefficients 3 and 21 are not divisible by nine, as required by [theorem 5.5](#).

References

- [1] S. Klavžar, U. Milutinović, and C. Petr, *Stern polynomials*, Advances in Applied Mathematics **39** (2007), 86–95. Author-hosted preprint: <https://users.fmf.uni-lj.si/klavzar/preprints/SternPolyFinal.pdf>.
- [2] M. Ulas, *Strong arithmetic property of certain Stern polynomials*, Publicationes Mathematicae Debrecen **96** (2020), 401–422. [doi:10.5486/PMD.2020.8683](https://doi.org/10.5486/PMD.2020.8683). The text and conjecture numbering checked in this note are those of [arXiv:1909.10844v1](https://arxiv.org/abs/1909.10844v1), 24 September 2019. The final journal PDF was not inspected.
- [3] D. Altizio, *Zeros of Stern polynomials in the complex plane*, [arXiv:2511.03847v1](https://arxiv.org/abs/2511.03847v1), 5 November 2025.